

# URIRECT

## Final App Evaluation Form

Version 1.0 | For 10 Registered Medical Technologists

|                   |                                                            |
|-------------------|------------------------------------------------------------|
| Evaluation target | URIRECT Android 1.3.0+4                                    |
| Visual model      | production_semiquant_knn_markerless_roi_topfix_v3_20260908 |
| UTI module        | uti_bayesian_lr_v1_1_20260926                              |
| Renal module      | renal_followup_rules_v1.1_20260919                         |
| Framework         | Software evaluation guided by ISO/IEC 25010                |

### Purpose

This form evaluates the final application's usability, functionality, reliability, observed performance, offline operation, and privacy controls. It is separate from the two-medtech review of Bayesian parameter values.

**This evaluation does not clinically validate the Bayesian posterior, establish diagnostic accuracy, approve treatment, or replace comparison with laboratory reference methods.**

### Confidentiality

Use an evaluator code. Do not write a patient name or another direct patient identifier on this form.

## Evaluator and Test Setup

| Field                              | Response |
|------------------------------------|----------|
| Evaluator code                     |          |
| Years of urinalysis experience     |          |
| Workplace / role                   |          |
| Phone model and Android version    |          |
| Strip manufacturer / catalog / lot |          |
| Test date                          |          |
| App version shown                  |          |

## Required Task Check

Mark Pass, Fail, or Not tested. Describe every failure or unexpected behavior.

| ID  | Task                                                                                                                                     | Pass                     | Fail                     | N/T                      | Comment |
|-----|------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|---------|
| T1  | Install and open the release APK.                                                                                                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |
| T2  | Complete a valid scan while airplane mode is enabled.                                                                                    | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |
| T3  | Confirm exactly ten analyte results are shown.                                                                                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |
| T4  | Confirm a blank or no-strip image is rejected.                                                                                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |
| T5  | Confirm a blurred image is rejected.                                                                                                     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |
| T6  | Confirm a partial or wrongly framed strip is rejected.                                                                                   | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |
| T7  | Confirm a low-confidence scan requests a retake.                                                                                         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |
| T8  | Complete the eligibility and symptom checklist.                                                                                          | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |
| T9  | Confirm an eligible UTI result shows its provisional posterior, prior, and applied factors separately from warnings and renal follow-up. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |
| T10 | Save, reopen, and delete a scan-history record.                                                                                          | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |         |

# App Quality Rating

Use one response per statement: 4 Strongly agree, 3 Agree, 2 Disagree, 1 Strongly disagree. Mark N/O only when the behavior was not observed; N/O is not a numerical score.

## A. Functional Suitability

| ID | Statement                                                                                                                             | 1                        | 2                        | 3                        | 4                        | N/O                      |
|----|---------------------------------------------------------------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| F1 | The app supports the complete capture-to-results workflow.                                                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| F2 | The app displays all ten URS-10T analyte results clearly.                                                                             | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| F3 | Invalid or uncertain scans lead to a retake instead of fabricated results.                                                            | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| F4 | UTI findings, systemic warnings, and renal follow-up are clearly separated.                                                           | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| F5 | The UTI estimate transparently shows its prior and applied factors, and the displayed actions match the findings entered or observed. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## B. Interaction Capability and Usability

| ID | Statement                                                            | 1                        | 2                        | 3                        | 4                        | N/O                      |
|----|----------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| U1 | The scan instructions are easy to understand.                        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| U2 | The capture controls and progress feedback are easy to follow.       | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| U3 | The symptom and safety questions are understandable.                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| U4 | Results can be read without confusing them with a diagnosis.         | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| U5 | Retake and consultation messages tell the user what to do next.      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| U6 | Text, controls, and result tables remain readable on the test phone. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## C. Reliability and Safety Behavior

| ID | Statement                                                                      | 1                        | 2                        | 3                        | 4                        | N/O                      |
|----|--------------------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| R1 | Repeating the same workflow produces stable app behavior.                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| R2 | The app recovers clearly from an invalid image or failed scan.                 | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| R3 | Serious reported symptoms remain visible even when the scan is invalid.        | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| R4 | The app avoids treatment or definitive-diagnosis wording.                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| R5 | Saved results retain the analytes, action, and rule details needed for review. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## D. Performance Efficiency

| ID | Statement                                                       | 1                        | 2                        | 3                        | 4                        | N/O                      |
|----|-----------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| P1 | The app opens within an acceptable time on the test phone.      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| P2 | Scan processing finishes within an acceptable time for RHU use. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| P3 | Navigation remains responsive during normal use.                | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

## E. Compatibility, Offline Operation, and Privacy Controls

| ID | Statement                                                         | 1                        | 2                        | 3                        | 4                        | N/O                      |
|----|-------------------------------------------------------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|
| C1 | The complete tested workflow works in airplane mode.              | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| C2 | Camera or gallery access behaves correctly on the test phone.     | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| C3 | Saved scans remain available after closing and reopening the app. | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| C4 | A saved scan can be deleted through the app.                      | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |
| C5 | The app avoids requesting information unnecessary for screening.  | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> | <input type="checkbox"/> |

Observed scan-processing time: \_\_\_\_\_ seconds

## Open Comments

Most useful feature:

Most important issue:

Suggested correction:

Any wording that could be misunderstood clinically:

## Overall Decision

☐ Acceptable for the stated thesis prototype scope.

☐ Acceptable after minor revisions.

☐ Major revision and repeat evaluation required.

☐ Not acceptable for the stated scope.

Evaluator signature: \_\_\_\_\_

Date: \_\_\_\_\_

## Researcher Scoring Instructions

1. Encode ratings as 1 to 4 and leave N/O missing.
2. Report mean and standard deviation for each item and quality section.
3. Report task success as passes divided by tasks attempted; exclude Not tested.
4. Report the count and percentage for each overall decision.
5. Preserve de-identified forms and the raw scoring sheet.
6. Do not invent a pass threshold after viewing results. Use an adviser-approved threshold declared before collection, or report descriptive results.
7. Describe this as an ISO/IEC 25010-guided evaluation, not ISO certification.